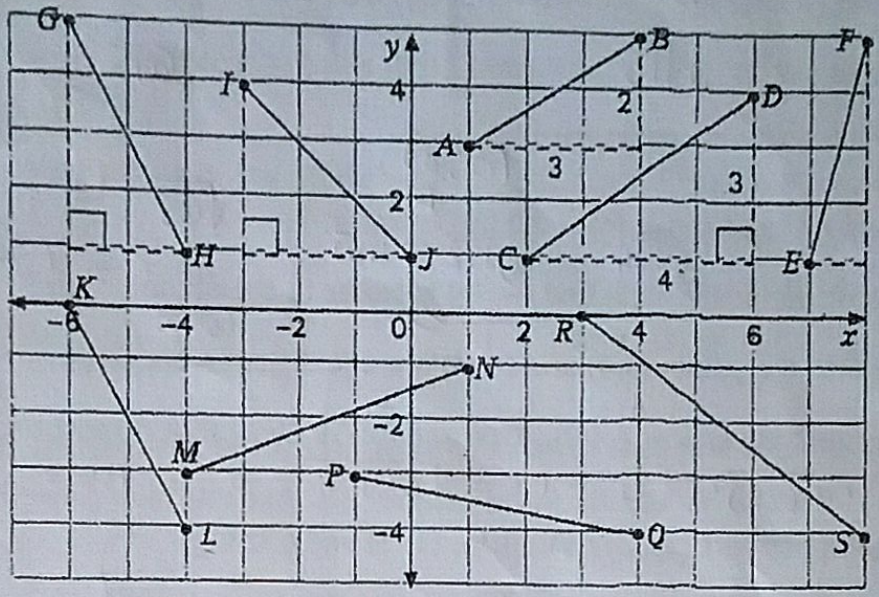
2. DISTANCE

Use Pythagoras' theorem to find the length of each line segment given (leave answers in root/ surd form)



AB = $\sqrt{13}$ CD= 5 EF = $\sqrt{17}$ GH = $2\sqrt{5}$

IJ = $3\sqrt{2}$ KL = $2\sqrt{5}$ MN = $\sqrt{29}$ PQ = $\sqrt{26}$

RS = $\sqrt{41}$ GC = HA = LE =

QF = LJ = IQ =

We can write Pythagoras' statement as follows and it becomes our "distance formula"

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

DISTANCE EXERCISE

1. Use the distance formula $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, to find the distance between each set of points:

a) (2;3) and (5;7) 5

b) (0;1) and (6;9) 10

c) (0;0) and (5;12) 13

d) (-4;8) and (2;0) 10

e) (-3;0) and (0;4) 5

f) (0;-1) and (2;-4) $\sqrt{13}$

g) (2;5) and (1;-3) $\sqrt{65}$

h) (4;9) and (-1;-3) 13

i) ((5;-3) and (-2;-7) $\sqrt{65}$

j) (2;-3) and (-2;3) $2\sqrt{13}$

k) (0;0) and (-2,5;6) 6,5

Answer on next page

2. a) Calculate the distance from the origin to (-4;-7) $\sqrt{65}$ (0;0)

b) Which of the points (3;5) and (-4;-4) is closer to the origin? $\sqrt{34}$ $\sqrt{32} = 4\sqrt{2}$ → (-4;-4) is closer

c) Show that the points (3;6) and (-1;4) are equidistant from (2;3). $\sqrt{10}$ for both

d) A triangle has vertices A(-4;0), B(0;3) and C(0;-3). Show that it must be isosceles.

$AB = \sqrt{(-4)^2 + (-3)^2} = 5$ $BC = \sqrt{0^2 + 6^2} = 6$ $AC = \sqrt{(-4)^2 + 3^2} = 5$

$AB = AC$

3. A quadrilateral has vertices (-1;-1), (3;1), (1;-3) and (-3;-5)

a) Find the length of its sides. What sort of quadrilateral is it?

b) Find the length of its diagonals.

} answer below

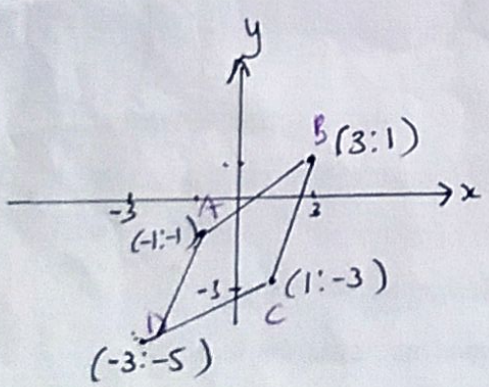
4. Prove that a circle can be drawn with centre (-2;1) and passing through the points (-5;3), (-4;-2) and (1;3). Show each radius = $\sqrt{13}$

Answer on next page

5. If Δ ABC is given, determine the perimeter of the triangle if A(1; -3) B(7; 3) C(4; 6).

answer below

3



a) $\sqrt{(-1-3)^2 + (-1-1)^2}$

$AB = 2\sqrt{5}$

$\sqrt{(-3-1)^2 + (-5+3)^2}$

$DC = 2\sqrt{5}$

$\sqrt{(3-1)^2 + (1--3)^2}$

$BC = 2\sqrt{5}$

$\sqrt{(-3--1)^2 + (-5--1)^2}$

$AD = 2\sqrt{5}$

Rhombus (all sides are =)

b) $AC = \sqrt{(-1-1)^2 + (-1--3)^2}$

$= 2\sqrt{2}$

$BD = \sqrt{(3--3)^2 + (1--5)^2}$

$= 6\sqrt{2}$

5) $AB = \sqrt{6^2 + 6^2}$

$= 6\sqrt{2}$

$BC = \sqrt{3^2 + (-3)^2}$

$= \sqrt{18}$

$= 3\sqrt{2}$

$AC = \sqrt{3^2 + 9^2}$

$= 3\sqrt{10}$

Perimeter = 22,21

$9\sqrt{2} + 3\sqrt{10}$

Page 5 Distance Exercise (Memo)

② a) Origin $(0;0) \rightarrow (-4;7)$ ($x_1\ y_1$ and $x_2\ y_2$)

$$\text{Distance} = \sqrt{(-4-0)^2 + (7-0)^2}$$
$$= \sqrt{16+49}$$
$$= \sqrt{65}$$

b) Origin $\rightarrow (3;5)$

$$\text{Distance} = \sqrt{(3-0)^2 + (5-0)^2}$$
$$= \sqrt{9+25}$$
$$= \sqrt{34}$$

Origin $\rightarrow (-4;-4)$

$$\text{Distance} = \sqrt{(-4-0)^2 + (-4-0)^2}$$
$$= \sqrt{16+16}$$
$$= \sqrt{32}$$

$\therefore (-4;-4)$ is closer to the origin

c) "Equidistant" means equal distance

$(3;6) \rightarrow (2;3)$

$$d = \sqrt{(3-2)^2 + (6-3)^2}$$
$$= \sqrt{1+9}$$
$$= \sqrt{10}$$

$(-1;4) \longrightarrow (2;3)$

$$d = \sqrt{(-1-2)^2 + (4-3)^2}$$
$$= \sqrt{9+1}$$
$$= \sqrt{10}$$

$\therefore$ Points $(3;6)$ and $(-1;4)$ are equidistant from point $(2;3)$

④ $(-2;1)$ to $(-5;3)$

$$r = \sqrt{(-2+5)^2 + (1-3)^2}$$
$$= \sqrt{9+4}$$
$$= \sqrt{13}$$

$(-2;1)$ to $(-4;-2)$

$$r = \sqrt{(-2+4)^2 + (1+2)^2}$$
$$= \sqrt{4+9}$$
$$= \sqrt{13}$$

$(-2;1)$ to $(1;3)$

$$r = \sqrt{(-2-1)^2 + (1-3)^2}$$
$$= \sqrt{9+4}$$
$$= \sqrt{13}$$

$\therefore$ all radii measure $\sqrt{13}$

i.e The 3 points lie on the circumference